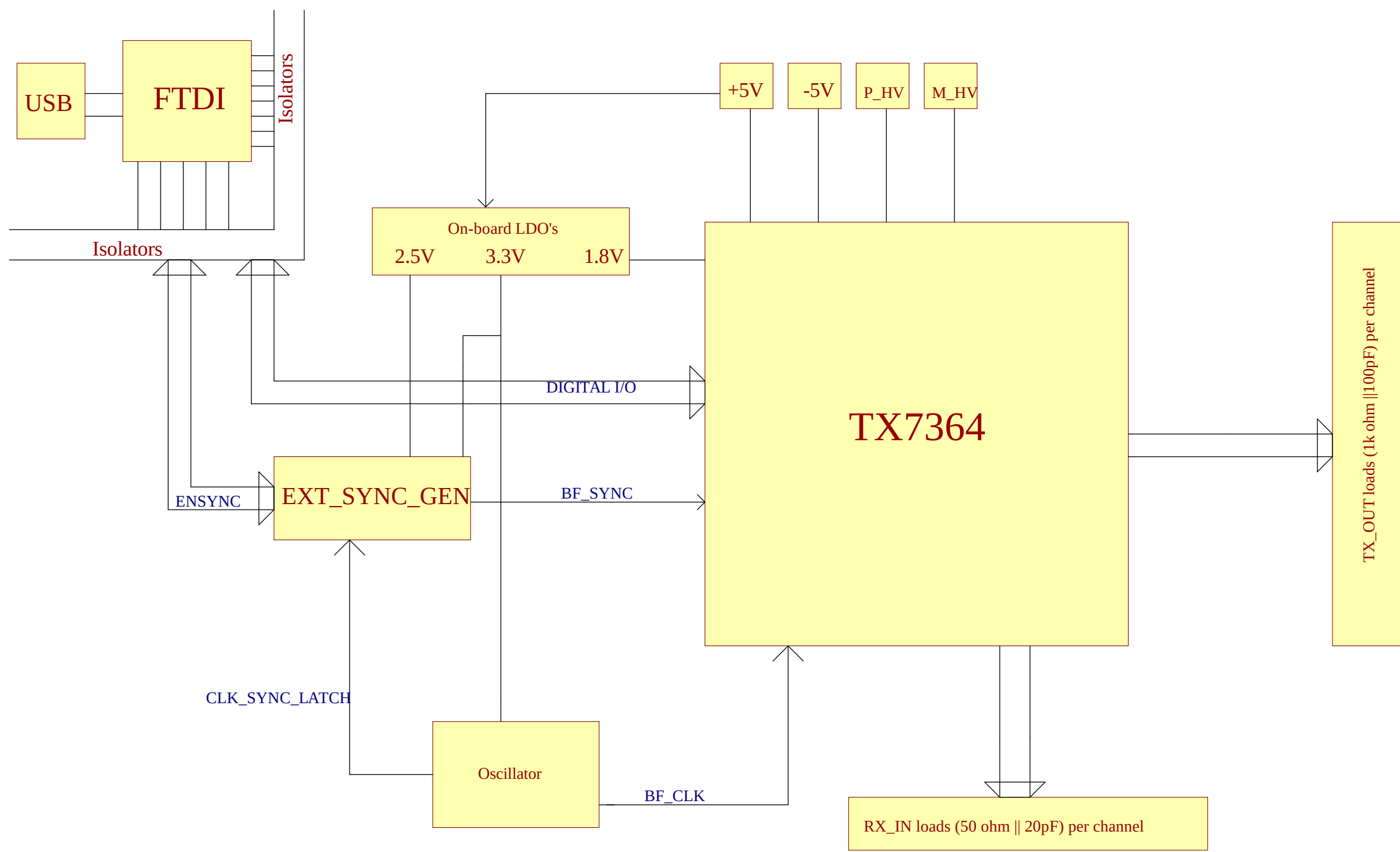


TX7364 Customer EVM : Functional block diagram



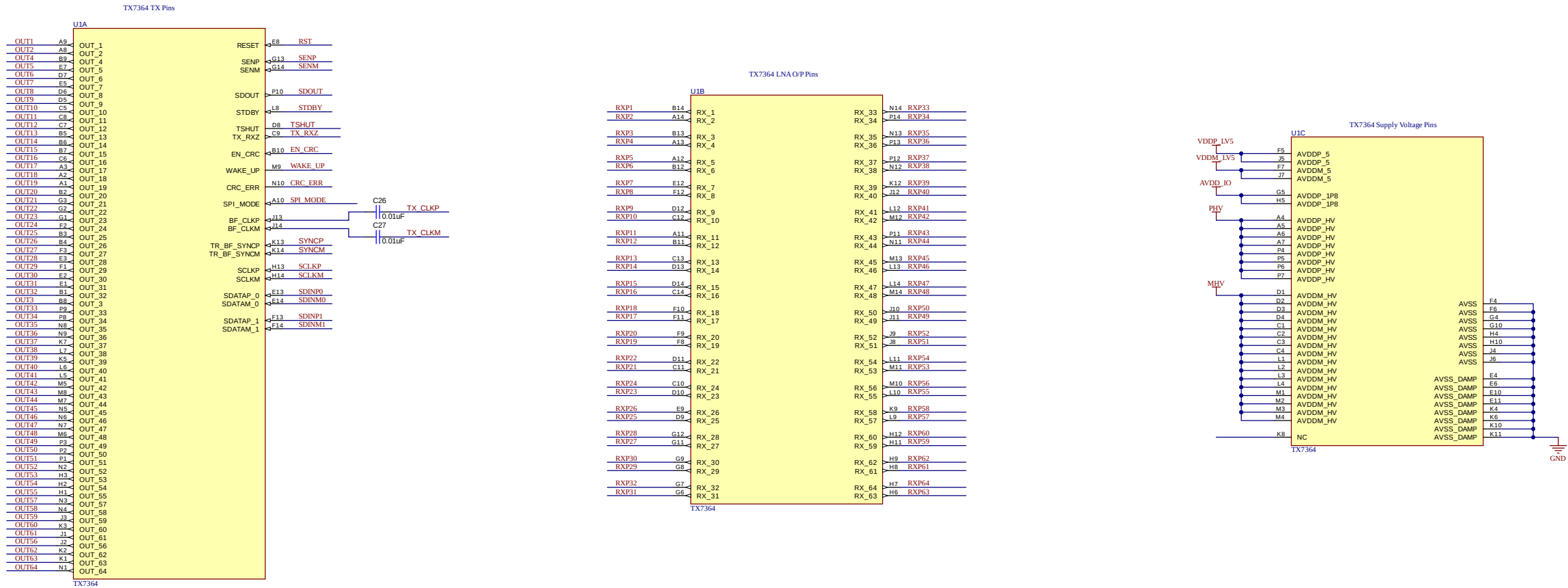
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Orderable: TX7364EVM	Designed for: Public Release	Mod. Date: 1/31/2024
TID #: N/A	Project Title: TX7364 EVM	
Number: DC309	Rev: A	Sheet Title: CoverPage
SVN Rev: Not in version control	Assembly Variant: Compact	Sheet: 1 of 11
Drawn By: KAVIKANNAN	File: DC309A_CoverPage_Compact.SchDoc	Size: C
Engineer: ROUSHAN	Contact: http://www.ti.com/support	http://www.ti.com

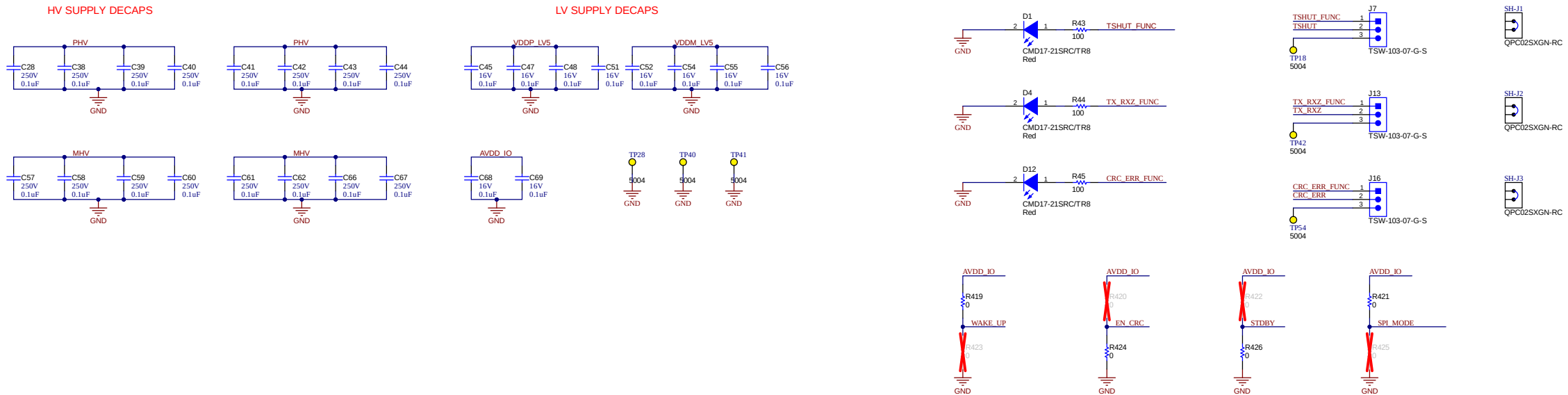


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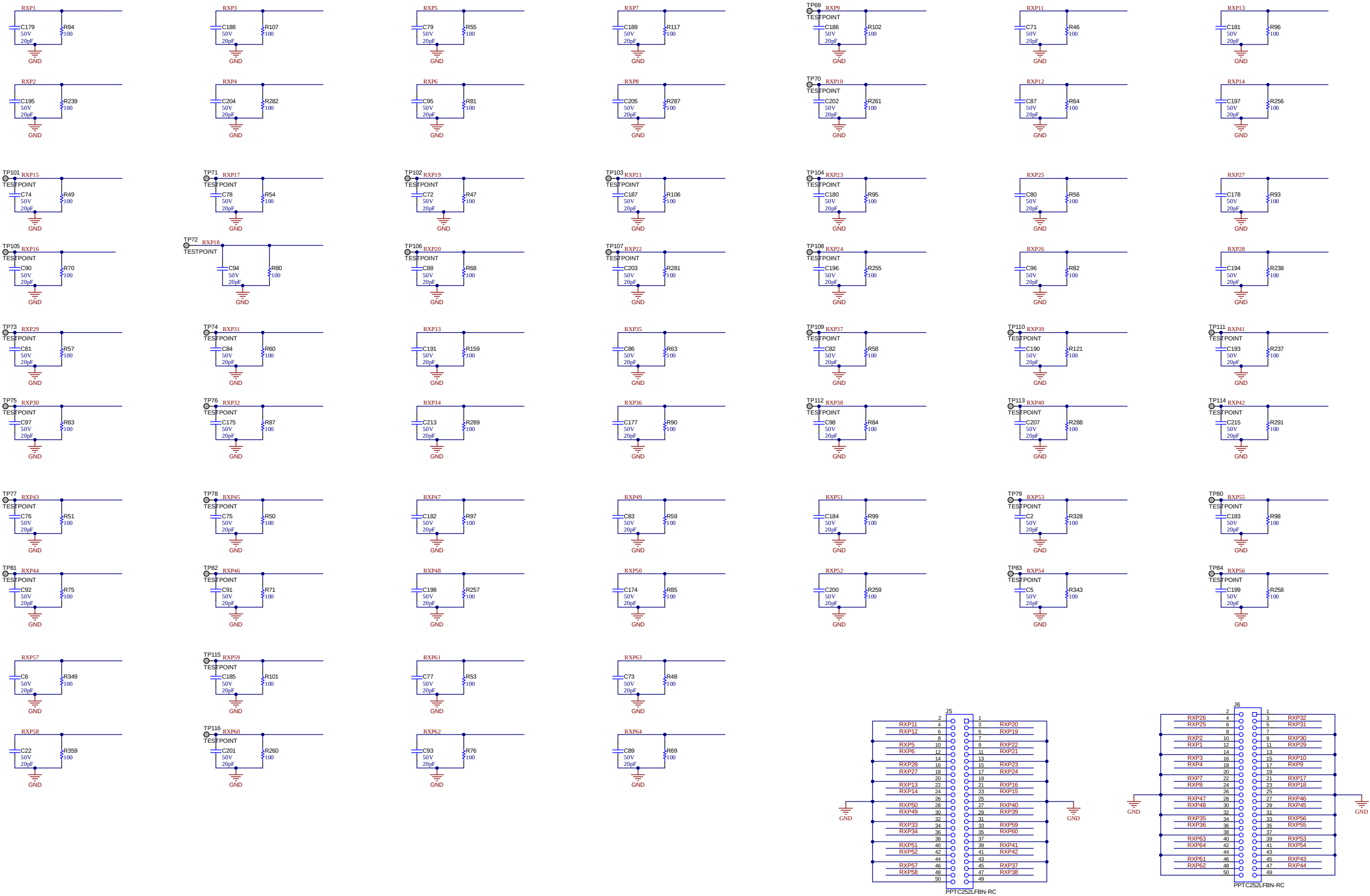
TX7364 IC PINOUT



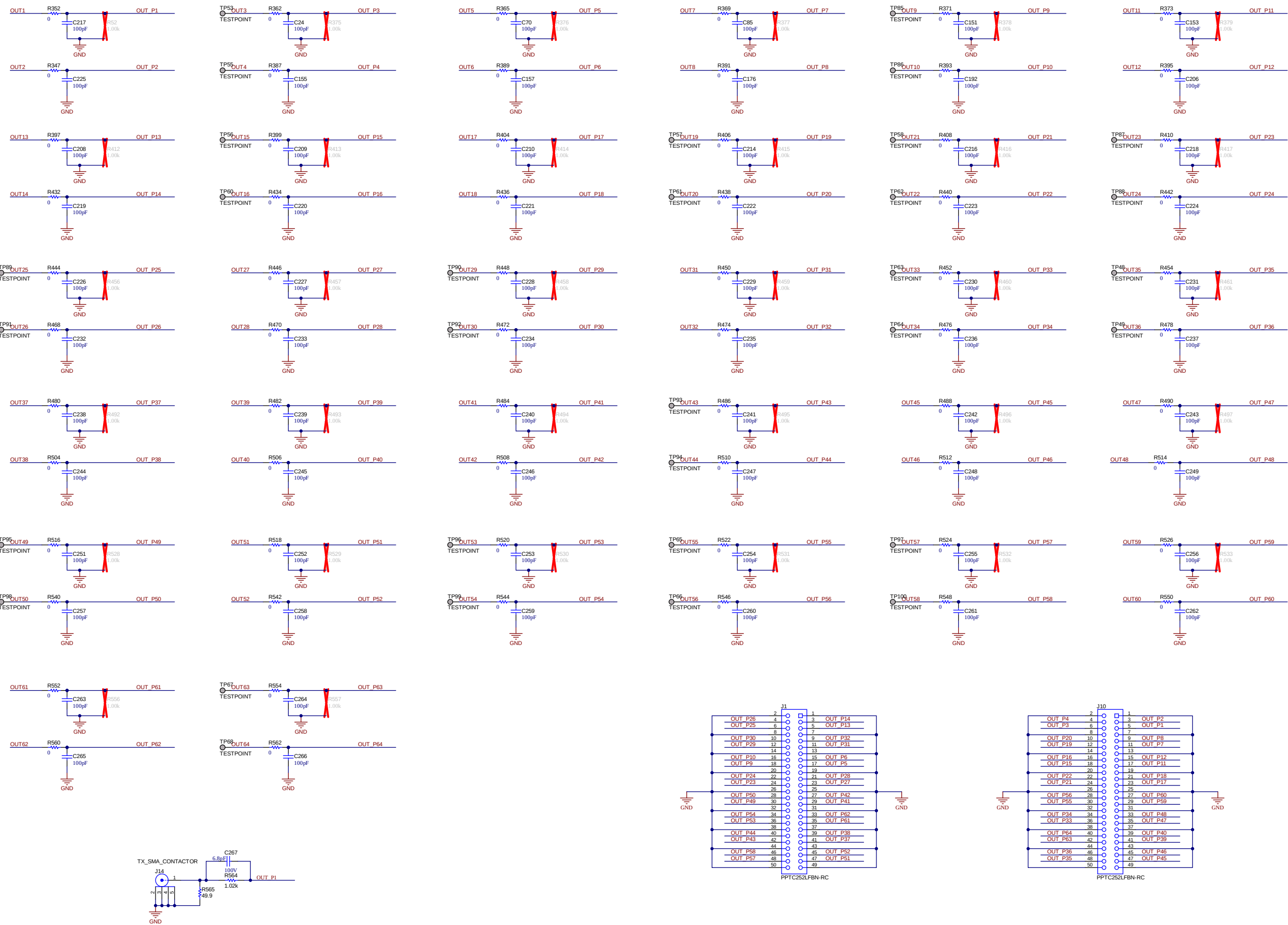
Digital Outputs



RX PINS AND LOADS



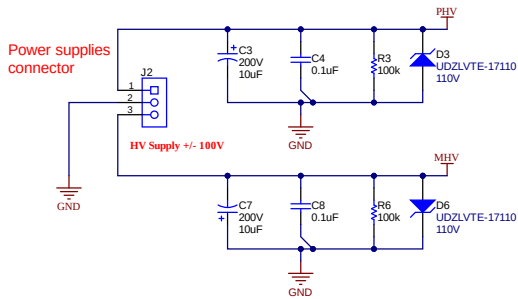
TRANSMITTER OUTPUT AND LOADS



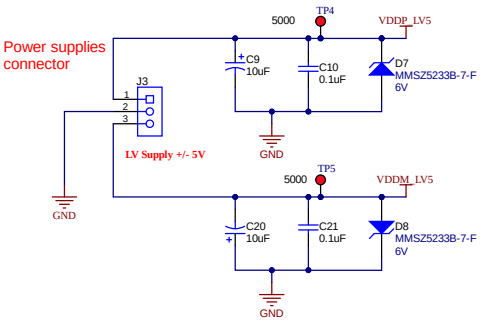
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Power Supplies and LDOs

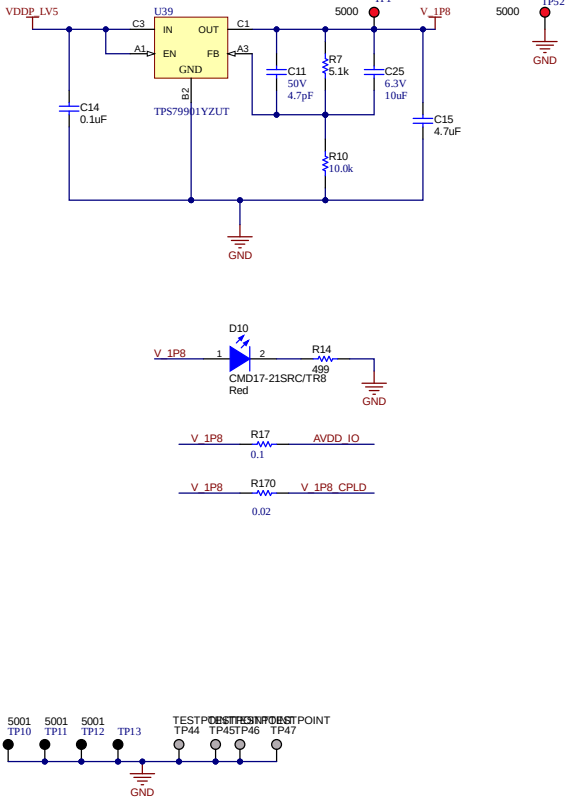
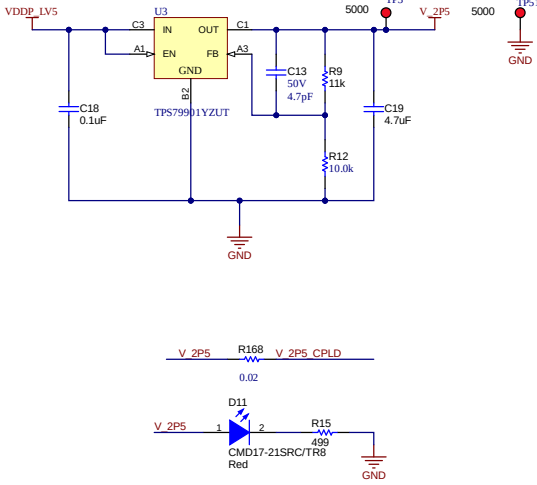
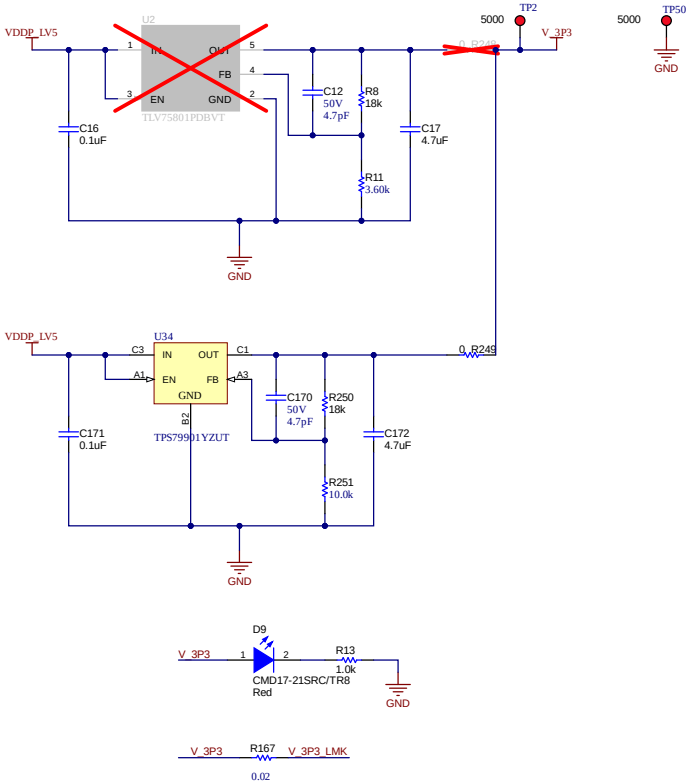
HV SUPPLIES



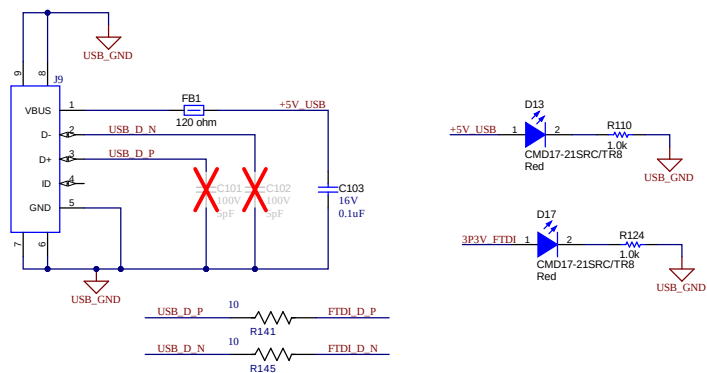
+5V SUPPLIES



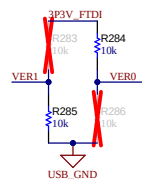
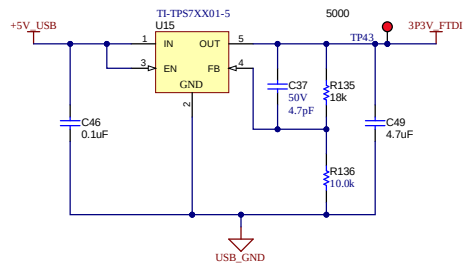
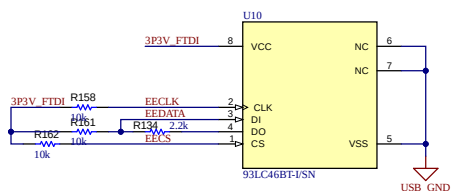
LDO'S TO GENERATE 3.3V, 2.5V AND 1.8V



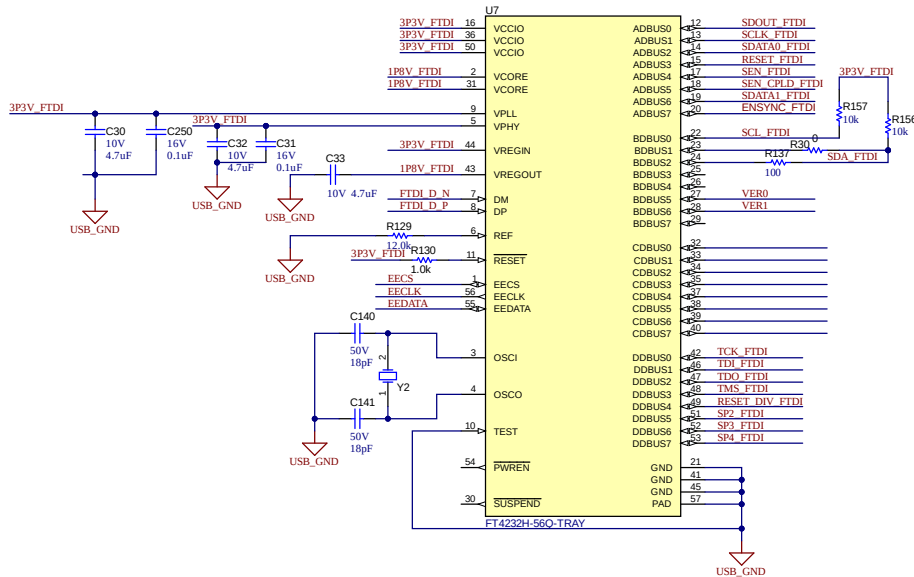
USB CONNECTOR AND FTDI PORT



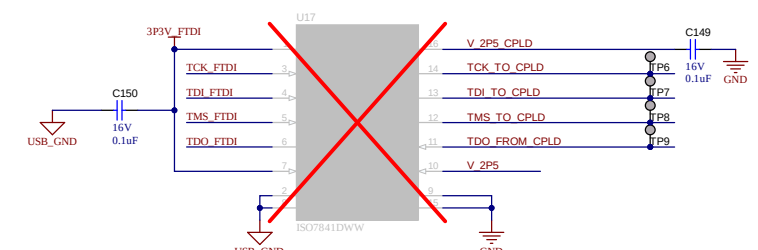
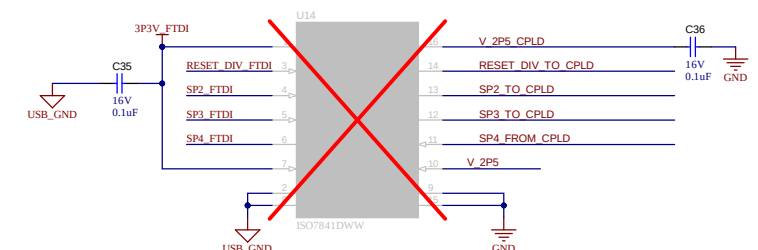
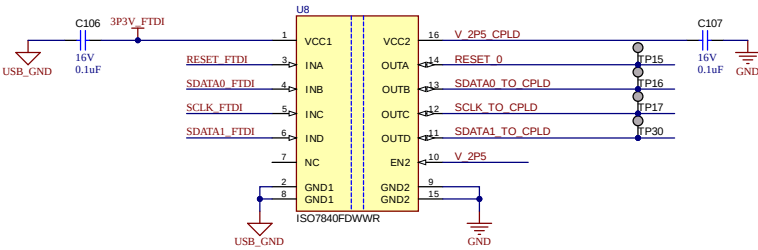
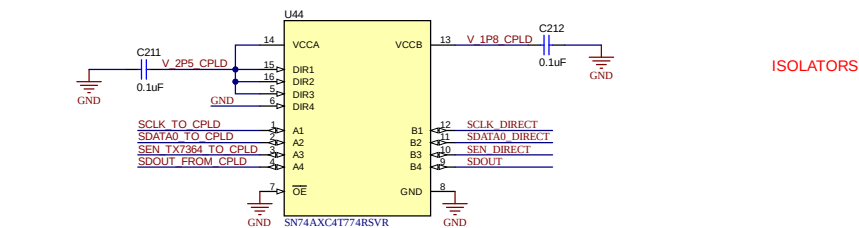
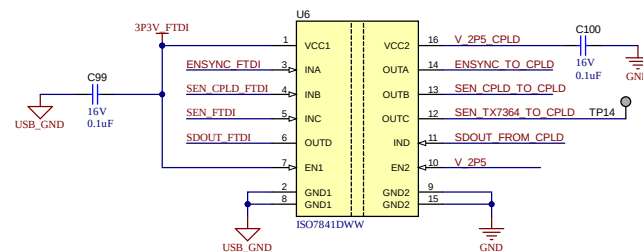
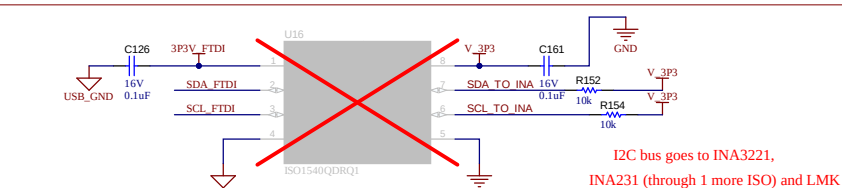
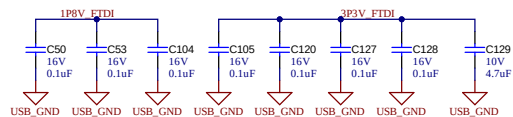
EEPROM



FTDI

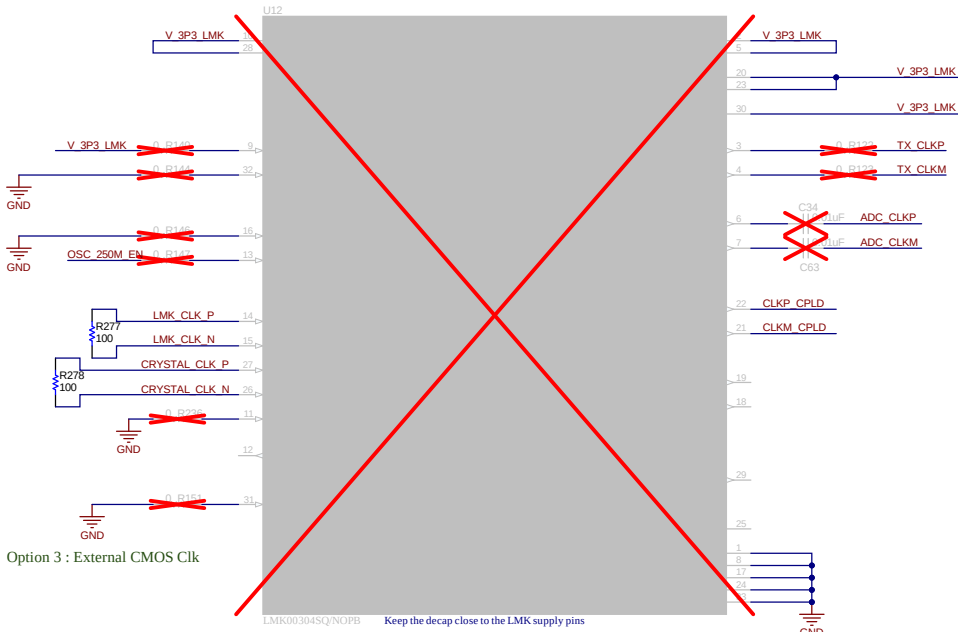


De Coupling caps (Place close to the IC pins)



CLK Generation

Clocking options in TX7364 EVM

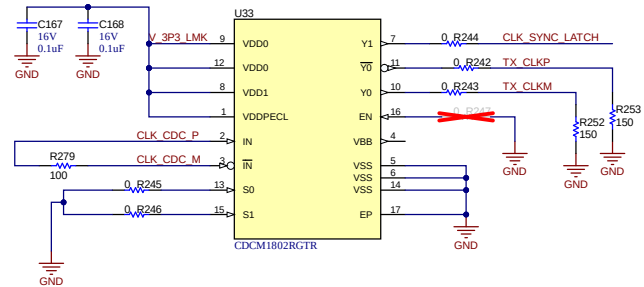
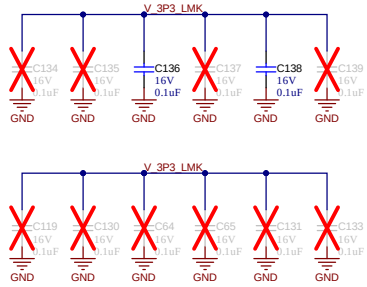
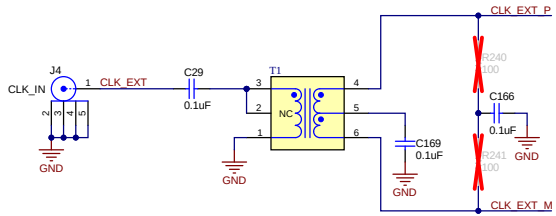
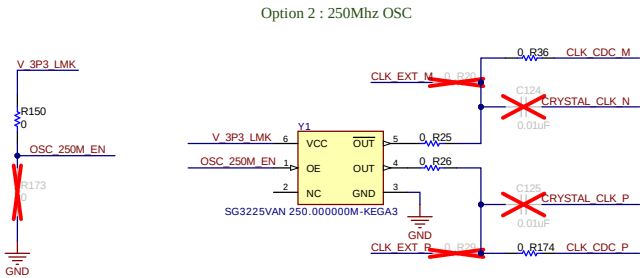


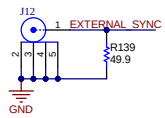
Clocking Option 1 : LMK61E2 (Only for CPLD version)
Install R148, R173 and uninstall R149, R150.
The Clock frequency can be programmed in TX7364GUL.

Clocking Option 2 : OSC_250MHz
Install R149, R150 and uninstall R148, R173.
The clock frequency is fixed to 250.0MHz.

Clocking Option 3 : External Clock
Install R149, R150, R20, R29 and uninstall R148, R173, R25, R26.
Provide a CMOS clock (0V->2.5V) to J4 SMA

External SYNC: Solder Resistor R276 and desolder R295
Provide CMOS SYNC signal to J13 (0V->2.5V)

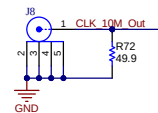
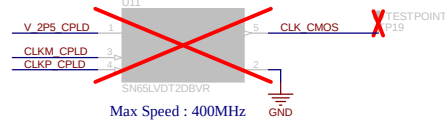




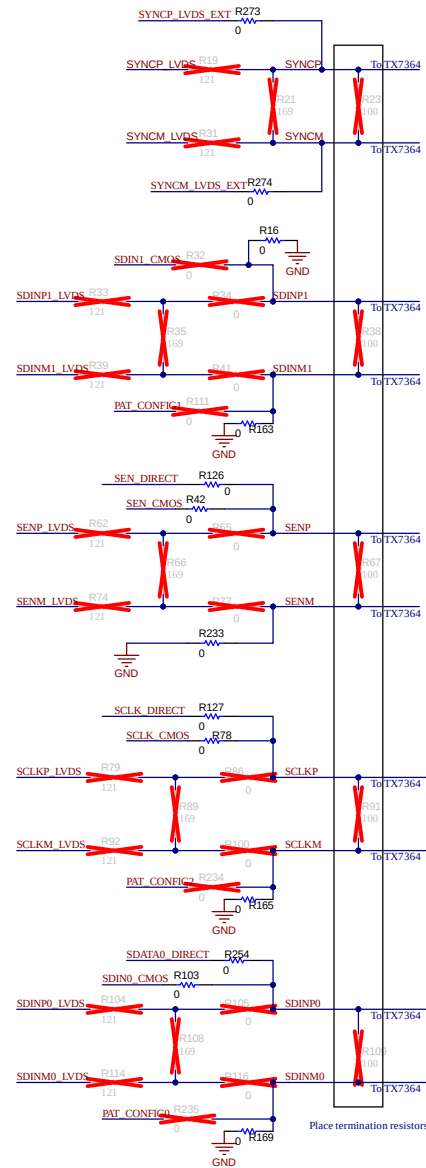
CPLD

CLOCK FOR CPLD (BF_CLK / 4)

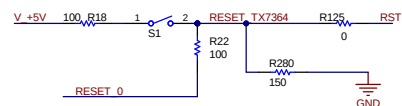
CPLD clock is inverted with respect to the BF_CLK



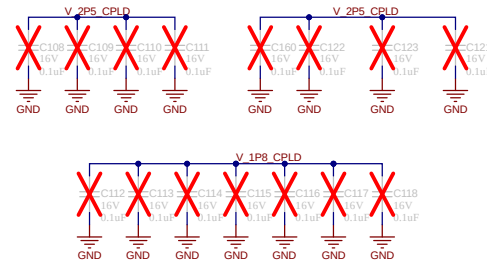
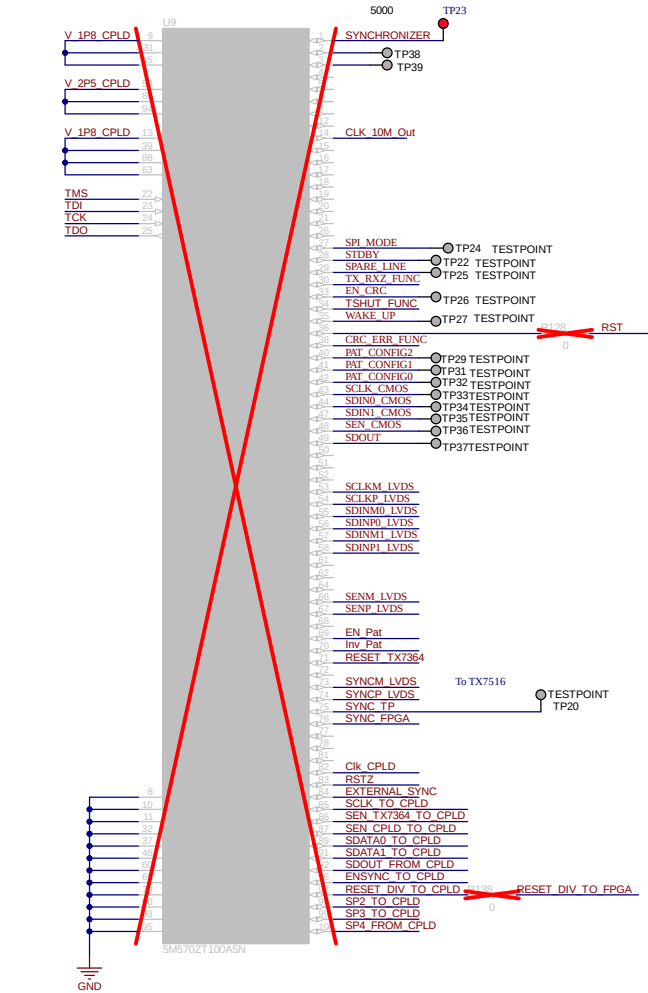
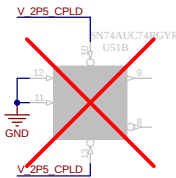
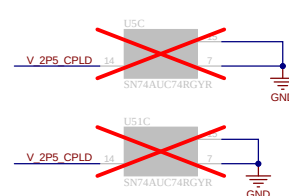
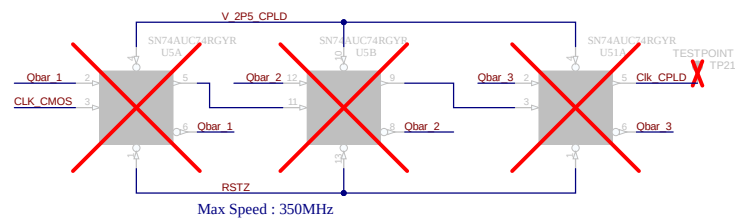
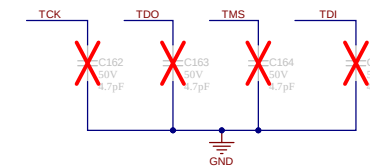
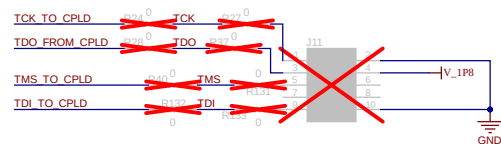
External resistor network for Emulated LVDS

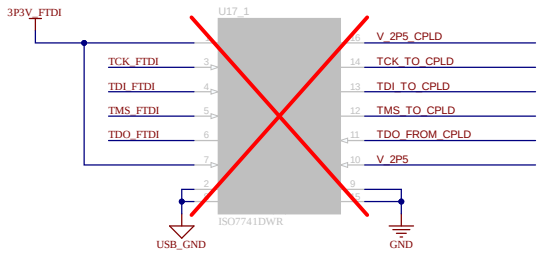
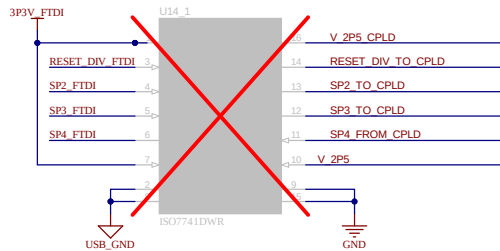
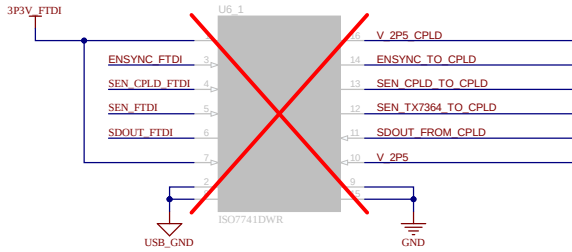
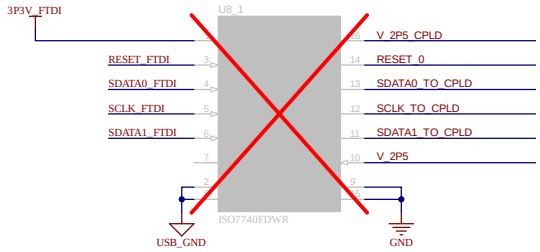


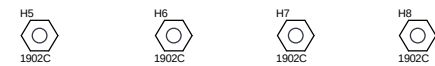
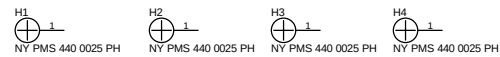
Place termination resistors close to TX7364



JTAG CONNECTOR







PCB Number: DC309
PCB Rev: A

PCB
LOGO
Texas Instruments



PCB
LOGO
FCC disclaimer

PCB
LOGO
WEEE logo



ZZ1
Assembly Note
These assemblies are ESD sensitive, ESD precautions shall be observed.

Assembly Note
These assemblies must be clean and free from flux and all contaminants. Use of no clean flux is not acceptable.

Z73
Assembly Note
These assemblies must comply with workmanship standards IPC-A-610 Class 2, unless otherwise specified.

H9
MECH
AK67421-0.3

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Orderable: TX7364EVM	Designed for: Public Release	Mod. Date: 2/5/2024	
ID #: N/A	Project Title: TX7364 EVM		
Number: DC309	Rev: A	Sheet: 11 of 11	
SVN Rev.: Not in version control	Assembly Variant: Compact		
Drawn By: KAVIKANNAN	File: DC309A_EVM_Hardware_SchDoc	Size: C	
Engineer: ROSHAN	Contact: http://www.ti.com/support		http://www.ti.com © Texas Instruments 2018